

Supplementary Material:

Every Fact, and how it was obtained

for *One Tile, Two Units: Area Inflation in $\mathbb{Q}(\sqrt{15})$ and Boundary Inflation in $\mathbb{Q}(\sqrt{5})$ for the Spectre Monotile*

Brent S. Hartshorn  (brenthartshorn@proton.me)

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What this is

The main paper stratifies its statements by epistemic weight: *Facts* are computations or theorems, *Propositions* carry one modelling step, *Conjectures* are sharp and unproved. This document lists every Fact — all 58 — with the key under which `spectrefacts.py` computes it, the claim in the form the paper states it, the first sentence of the method by which it was obtained, and a mark on the 18 Facts that are also proved by the Lean 4 kernel.

Three things about the table.

- **It is generated, not written.** Every row is emitted at build time from `spectrefacts.py`, the same module that produces every number in the main paper. There is no second copy to drift.
- **The Lean column is measured, not asserted.** A checkmark means the Fact's `lean` field names a theorem that exists in `SpectreFacts.lean` — 51 theorems, kernel-accepted, no admitted proofs — and the paper's build fails if a named theorem is missing. Six such phantom attributions were found and removed when that check was added; the count of checked Facts fell from twenty-two to eighteen, and eighteen is the honest number.
- **The method column is the first sentence.** A method that runs to a paragraph is truncated at its first full stop or semicolon. The full text is in the module, one lookup by key away.

Two of the Facts are inputs rather than results, and the paper says so: `charpoly` rests on the transcribed substitution matrix, and `boundary_recurrence` on a perimeter seed counted from placed tiles. Both are audited by an independent route — the matrix against a second implementation of the same substitution, the seed against an exact $\mathbb{Q}(\sqrt{3})$ recount — and an audit is not a proof. They are the two rows a sceptical reader should look at first.

The table

Generated from `spectrefacts.py`. The last column marks the Facts that are also proved in Lean, see <https://github.com/brentharts/spectre/wiki/Spectre-Lean>; the others are computations.

Key	Claim	How	Lean
<code>charpoly</code>	$\chi_M(x) = x^5(x-1)(x+1)(x^2-8x+1)$	exact characteristic polynomial over $\mathbb{Z}$, factored	
<code>perron</code>	$\lambda^2 = 4 + \sqrt{15}$	largest root of the quadratic factor	
<code>pell</code>	$4^2 - 15 \cdot 1^2 = 1$	the Pell equation, so λ^2 is the fundamental unit of $\mathbb{Z}[\sqrt{15}]$	✓
<code>lam_closed</code>	$\lambda = \frac{1}{2}(\sqrt{6} + \sqrt{10})$	square both sides: $(\sqrt{6} + \sqrt{10})^2/4 = (16 + 4\sqrt{15})/4$	
<code>minpoly</code>	$x^4 - 8x^2 + 1$	minimal polynomial of λ over $\mathbb{Q}$	
<code>channel_hex</code>	$(\lambda - \lambda^{-1})^2 = 6$	$\lambda^2 + \lambda^{-2} - 2$, in $\mathbb{Z}[\sqrt{15}]$, no root evaluated	✓
<code>channel_pent</code>	$(\lambda + \lambda^{-1})^2 = 10$	$\lambda^2 + \lambda^{-2} + 2$, in $\mathbb{Z}[\sqrt{15}]$, no root evaluated	✓
<code>contraction</code>	$\lambda^{-4} = 31 - 8\sqrt{15}$	the square of the conjugate unit	✓
<code>jordan</code>	$\text{rank } M = 5$, $\text{rank } M^n = 4$ ($n \geq 2$)	exact elimination over $\mathbb{Q}$	

Key	Claim	How	Lean
q_plus_eigen	$Q_+ M = Q_+$	left eigenvector for the marginal eigenvalue +1	✓
q_minus_eigen	$Q_- M = -Q_-$	left eigenvector for the marginal eigenvalue -1	✓
q_values	$Q_+ \equiv -1, \quad Q_- \equiv +1, -1, +1, -1$	evaluated on the matrix census at depths one to four	✓
censuses	$T = 1, 8, 63, 496 ; T_{\text{geom}} = 1, 9, 71, 559$	the same patches counted with and without the Mystic	
mystic_rule	$T_{\text{geom}}(n) = T(n) + T(n-1)$	row Gamma of M is all ones, so N_Gamma(n) is the whole census at n-1 and the Mystic is a second copy of it	✓
ledger	$N_{\Theta}(n) = N_{\Gamma}(n-1)$	row Theta of M is the indicator of Gamma, so the count is the previous generation of Gamma verbatim	
ledger_lambda	$N_{\Lambda}(n) = N_{\Sigma}(n-1)$	row Lambda of M is the indicator of Sigma	
hat_charpoly	$\chi_H(x) = (x-1)(x+1)(x^2-7x+1)$	characteristic polynomial of the four-metatile Hat substitution	
hat_perron	$\kappa = \frac{1}{2}(7+3\sqrt{5}) = \varphi^4$	largest root of the Hat quadratic, and the fourth power of the golden ratio	
phase_fields	$\mathbb{Q}(\sqrt{15})$ against $\mathbb{Q}(\sqrt{5})$	the two characteristic polynomials differ in a single coefficient, seven against eight, and that...	
frequencies	$v \in \mathbb{Z}[\sqrt{15}]^9, \sum v = 1$	nullspace of M - lambda^2 I, normalised	✓
hat_frequencies	$f_H = \frac{1}{3} ; f_T, f_P, f_F \notin \mathbb{Q}$	Perron eigenvector of the four-metatile Hat substitution, over $\mathbb{Q}(\sqrt{5})$	
aperiodicity_witness	9/9 vs 3/4 irrational	a periodic tiling has rational frequencies, so an irrational one refutes periodicity	✓
trace_accounting	tr = unit + conjugate : 8, 7	the marginal pair +1 and -1 cancels and the zero modes contribute nothing, so the trace is the unit plus...	✓
minors	$D_m = (-1)^m(m+1)$	leading principal minors of the tridiagonal (-2,1,1) form, in closed form	✓
signature	$\sigma(T(2,k)) = -(k-1),$ $u(T(2,k)) = \frac{k-1}{2}$	the minors alternate in sign, so the form is negative definite by Sylvester	
area_form	$A(a,b) = 2\sqrt{3}a^2 + 3ab + \sqrt{3}b^2$	symbolic shoelace on the published vertex construction, an identity in $\mathbb{Q}(\sqrt{3})[a,b]$ rather than a fit to...	
area_endpoints	$A(1, \sqrt{3}) = 8\sqrt{3}, A(\sqrt{3}, 1) = 10\sqrt{3}$	the Hat and Turtle endpoints of the Tile(a,b) family	
mystic_asymmetry	$A(b,a) - A(a,b) = \sqrt{3}(b^2 - a^2)$	the exact content of "only Gamma moves": the area form is not symmetric, and the Mystic is the species...	
order_parameter	$\max f_{\Gamma} - f_{\Delta} = \frac{35}{67} - \frac{20}{201}\sqrt{15}$	the splitting maximum at the Hat endpoint	
rho_x	$\rho_X(p)$ with $p = \Pr(\Phi^2)$ the only unknown	bulk X-density from the per-species X-charges	
x_charge	$X_{\Gamma_2} = 14 ; X_{\Gamma_1} = X_{\Delta} = X_{\Sigma} = 4 ;$ $X_{\Lambda} = 2 ; X_{\Phi} \in \{0, 2\}$	built the tiling and counted, for every interior tile, the shared edges whose two canonical slot roles...	✓
atlas	131 contact classes, stable	the unordered adjacency classes (species and slot on each side) number 131 at depths three, four, five and...	
phi_drift	$p = \Pr(\Phi^2) : 0.683, 0.579, 0.538, 0.519$	the flavour split by depth three to six	
rho_x_at_half	$p = \frac{1}{2} \implies \rho_X = \frac{2g}{1+g} = 1 - \frac{\sqrt{15}}{5}$	conditional, not established: repeated Aitken extrapolation of the depth three to six values gives 0.5107...	
chiral_angle	$\cos \theta_{\text{chiral}} = \frac{1}{8}(1+3\sqrt{5})$	Moritake et al	
angle_from_kappa	$\cos \theta_{\text{chiral}} = \frac{1}{4}(\kappa-3)$	3 phi - 1 and phi^4 - 3 are the same number, so the measured chiral angle is a function of the Hat...	✓
spectre_angle_differs	$(\lambda^2 - 3)/4 > 1$	substituting the Spectre inflation factor into the Hat formula gives a cosine greater than one, so the...	
bh_theorem	$u(7_1 \# \overline{7_1}) \leq 5 < 6 = u(7_1) + u(\overline{7_1})$	Brittenham and Hermiller 2025, arXiv:2506.24088, Theorem 1.2: the first failure of additivity of...	
bh_version	cite v2, not v1	the first version routed through two diagrams of K15n81556 asserted to be the same knot	
bh_gap_was_chirality	the gap was a mirror confusion	the two diagrams differed by mirroring, which is exactly the distinction the binding mechanism turns on	

Key	Claim	How	Lean
bh_deficit_bound bh_threshold	$1 \leq \delta \leq 4$ $k, \ell \geq 7$ odd	the deficit is bounded, not determined their Corollary 1.3 covers $T(2, 2k+1)$ with $k \geq 3$, that is index seven and above	
beta	$\beta_{\text{pred}} = \frac{70}{67} - \frac{40}{201}\sqrt{15}$	the order-parameter maximum read in degrees under the stated unit postulate – a modelling step, not a...	
beta_tension	$ \beta_{\text{pred}} - \beta_{\text{obs}} /\sigma$	against the 2026 joint ACT DR6 + Planck PR4 value 0.277 ± 0.057 degrees	
boundary_recurrence	$P(k) = 5P(k-1) - 3P(k-2) - P(k-3)$	exact integer recurrence on the supertile perimeter, derived in spectre_boundary_operator.py from the...	✓
boundary_chi	$\chi_{\partial}(x) = (x-1)(x^2 - 4x - 1)$	characteristic polynomial of the companion matrix of the recurrence	
boundary_growth boundary_field	$\nu = 2 + \sqrt{5} = \varphi^3$ $\nu \in \mathbb{Q}(\sqrt{5})$, $\lambda^2 \in \mathbb{Q}(\sqrt{15})$	Perron root of the boundary operator area and perimeter of the same tile inflate in different quadratic fields	
boundary_dimension	$\dim_{\text{box}} \partial = \log \nu / \log \lambda$	nu exceeds lambda, so the supertile boundary is fractal and the perimeter does not scale with the linear...	
balanced_exponent	$s^* = 1 - \log \nu / \log \lambda^2$	the thickness exponent making every substitution level carry comparable volume	
compositum	$\mu = \lambda^2 / \nu$ has degree 4	the balanced ratio lies in $\mathbb{Q}(\sqrt{3}, \sqrt{5})$ and in neither quadratic field alone, so the Spectre and the...	
golden_absent	$(x^2 - 4x - 1) \nmid \chi_M, \chi_H$	the boundary quadratic divides neither substitution matrix	
lucas_deflation	$\varphi^n : x^2 - L_n x + (-1)^n$	every power of the golden ratio has a Lucas quadratic for its minimal polynomial, so agreement of two such...	
spectre_not_golden	$\lambda^2 \neq \varphi^n$	the trace of the Spectre unit is eight, which is not a Lucas number, so the area inflation is not golden...	
boundary_invariant	$P(k+2) - 4P(k+1) - P(k) = -16$	the quadratic factor of chi applied to the sequence annihilates the growing part and leaves the constant...	✓
boundary_growth_bound	$P(k+3) \geq 4P(k+2) + 30$	the perimeter more than quadruples at every step past the seed, while lambda $\nmid 4$	✓
lam_below_four	$\lambda^2 < 16$	$16 - \lambda^2 = 12 - \sqrt{15}$, positive since $12^2 \nmid 15$	✓
boundary_exact	$P(k)$ counted exactly in $\mathbb{Q}(\sqrt{3})$	the same sequence recounted with edges identified by equality of exact quadratic-field coordinates rather...	